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The influence of heat treatment process on electroplated metallized through glass vias (TGV) substrates

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Highlights

- An in-depth discussion of the grain boundary evolution in copper pillars within the electroplated metallized through glass vias (TGV) substrates has been conducted.
- The applicability conditions for calculating residual stress using the curvature method and the $\sin^2\Psi$ method were discussed.

- The wear resistance data and resistivity of the copper pillars inside the vias have been calculated separately, and the results have been discussed in detail.
- Theoretical support has been provided for the adoption of heat treatment processes that meet the requirements for low warpage.

Abstract

Ensuring low warpage for electroplated metallized through glass vias (TGV) substrates constitutes a pivotal technical bottleneck in engineering applications. The regulation of residual stress in the electroplated layer to improve substrate warpage has been proven to be an effective strategy recently. In this study, we successfully improved the warping phenomenon of TGV substrates after metallization by employing an appropriate heat treatment process. Molecular dynamics simulation analysis indicates that the presence of titanium atoms can effectively promote the segregation behavior of copper on the SiO_2-Ti-Cu structure, enhancing interface stability. The applied heat treatment process promotes atomic diffusion, which can further improve the interface bonding strength between copper and SiO_2. During heat treatment, the warpage curvature radius of copper-overburden TGV substrates initially increases then decreases, peaking at 979 m at 100 ℃, which minimizes wafer warpage. However, the Vickers hardness (HU) of the interconnect copper pillars is the lowest at this temperature, with a value of 3.64×10^{-2} GPa. After heat treatment, electron backscatter diffraction (EBSD) reveals that electroplated copper pillars within glass vias exhibit grain refinement, with the 200 ℃ treated sample showing a high proportion of grain boundary misorientation angles (greater than 50°) up to 51.63 % and a minimum grain orientation spread (GOS) of 0.5, aiding crack resistance and reducing lattice distortion. When the temperature increases to 300 ℃, combined constrained thermal stress-induced twinning phenomena along the via axis direction occur, with the proportion of ∑3 and ∑9 type twin boundaries increasing to 45.46 %. The resistivity of the copper pillars increases gradually from 15.83×10^{-6} Ω·m at room temperature to 18.11×10^{-6} Ω·m after heat treatment at 300 ℃. X-ray photoelectron spectroscopy (XPS) analysis suggests that residual organic polymer additives are likely the main cause of the significant difference in resistivity between the electroplated copper pillars and oxygen-free copper. Overall, this study offers novel insights into fabricating low warpage electroplated metallized TGV substrates, balancing mechanical and electrical performance of vertical interconnect copper pillars.

Keywords

TGV substrate; Microstructure; Warping; Electrical property; Nano-indentation

1. Introduction

RF (Radio Frequency) chips are one of the core components of modern communication systems and are widely used in applications such as mobile phones, wireless network devices, satellite communications, and radar systems [1], [2], [3], [4]. With the development of 5 G and IoT (Internet of Things) technologies, the performance requirements for RF chips are increasingly stringent, and their 3D packaging technology has become a research focus [5], [6]. The 3D packaging of RF chips can enhance signal transmission speed, reduce volume, improve system performance, and achieve high-level integration [7]. In the 3D packaging of RF chips, the packaging substrate based on TGV (Through Glass Via) technology plays a crucial role. TGV technology achieves vertical interconnects of electrical signals within the glass layers by creating micro-holes in the glass substrate and filling them with conductive materials, providing a high-density, low-loss interconnect solution for RF chips [8], [9]. However, warpage of the substrate is a significant challenge in 3D packaging technology. During the metallization process of the TGV substrate, warping can easily occur due to the mismatch of thermal expansion coefficients between materials and the stress accumulation during the electroplating process, affecting the yield and processing precision of the grinding process, and thus impacting the overall performance and reliability of the RF chip [10], [11].

To address the warpage issue of wafers after metallization, researchers have explored various approaches. In recent years, methods of regulating the residual stress in the surface plating layer to improve wafer warpage have shown great research potential. Wang et al. [12] found that the residual stress in copper thin films can be reduced by increasing the electroplating temperature. At a temperature of 40 °C, the randomly oriented electroplated Cu films exhibit the lowest residual stress. Films with low residual stress can prevent the substrate from warping and reduce the failure rate in processes such as direct copper bonding and other alignment-dependent processes. Similarly, Cheng et al. [13] reported that electroplated layers containing nano-twinned copper (nt-Cu) can be obtained by using appropriate electroplating parameters, which not only effectively reduces wafer warpage but also decreases thermal stress through stress relaxation during subsequent thermal cycling tests. It is worth noting that the quality of the electroplating process is highly susceptible to equipment, materials, and the influence of the previous process during mass production, making it difficult to achieve the ideal electroplating conditions. That is to say, during mass production of the electroplating process, it is challenging to form an ideal low-residual-stress electroplated layer. Nt-Cu films have excellent thermal stability and yield stress, showing promising applications in inhibiting wafer warpage [14]. Chen et al. [15] found that when the thickness of nt-Cu film deposited on silicon wafers is 3 μm, it has the highest orientation transition temperature and the best thermal stability. However, significant challenges have been encountered when applying this process to the metallization of glass substrates with through-holes. This is because the electroplating filling of the through-holes is the primary objective, and the thickness of the electroplated copper layer on the glass surface mainly depends on the filling time of the through-holes. Generally, after the glass through-holes are filled, the thickness of the electroplated copper layer is above 20 μm, which adversely affects the regulation of residual stress using nt-Cu films. Therefore, it is necessary to develop a controllable and efficient technology to regulate the warpage of TGV substrates.

Heat treatment, as a key process in the semiconductor industry, plays an important role in controlling the warpage of TGV substrates [16], [17]. By conducting heat treatment under protective atmosphere or vacuum conditions, oxidation can be effectively reduced, the purity and density of copper films can be improved, and the movement of copper atoms within the film can be promoted, thereby optimizing their microstructure. Studies have shown that the heat treatment temperature is a key factor affecting the residual stress in electroplated copper films [18], and its influence is more significant than that of the film thickness or microstructure of the copper film [19]. Generally, with the implementation of heat treatment processes, the residual stress in copper overburden films can be redistributed to a certain extent, thereby affecting the warpage height of the glass substrate [20]. Under appropriate heat treatment process parameters, the warpage of the glass substrate can be controlled to an optimal state to meet the requirements of mass production yield. Although theoretically, the smaller the warpage height, the easier the processing of the copper overburden glass wafer in the chemical mechanical polishing (CMP) process. However, for glass wafers with through-holes already filled with copper, the suitable heat treatment parameters required not only depend on the warpage changes of the glass substrate but also on the electrical performance changes of the copper pillars that play a key role in vertical interconnect within the through-holes. As heat treatment progresses, changes in grain size can alter the material's microstructure, thereby affecting electron mobility and scattering mechanisms, which are key factors in resistivity changes [21]. Larger grain sizes may lead to reduced electron scattering, thus decreasing resistivity; whereas smaller grain sizes may increase scattering, resulting in increased resistivity. Therefore, changes in grain size are directly related to the resistivity characteristics of the copper pillars. Tan et al. [22] found that by providing a gradient thermal field, the annealing treatment enhanced the driving force for grain boundary evolution, resulting in copper films with a larger average grain size, better recrystallization rate, and higher average grain aspect ratio, significantly reducing the resistivity of the Cu films. During heat treatment, the grain growth patterns, grain boundary migration, and dislocation movement will differ under different processes, and these factors collectively determine the microstructure of the through-hole copper pillars (hereinafter referred to as TGV-Cu). Therefore, to explore the optimal combination of warpage height and TGV-Cu resistivity for copper overburden glass substrate, it is necessary to delve into the impact of temperature on the microstructural evolution of TGV-Cu and further elucidate the mechanism by which it affects resistivity changes through grain growth.

In this study, the warpage height of the copper overburden glass substrate was controlled by implementing a heat treatment process under a nitrogen-protected atmosphere. The evolution of warpage height, the type of TGV-Cu grain boundaries, the microstructural changes of grains, as well as variations in resistivity and micro-nano mechanical properties of the copper electroplated glass substrate at different temperatures were systematically investigated. Moreover, through molecular dynamics simulation, a theoretical analysis of the interface reliability between Cu and glass was conducted. This research provides a new process route and theoretical support for manufacturing low warpage electroplated metallized TGV substrates and optimal matching between mechanical and electrical performance of vertical interconnect copper pillars through heat treatment processes.

2. Experiment

In this study, quartz glass substrates with a thickness of approximately 330 μm and a side length of 200 mm were obtained from a commercial supplier. The raw material was processed into 50 mm*50 mm square glass pieces using laser machining. Subsequently, vertical through-holes were fabricated on the glass pieces by combining laser-induced methods with wet etching. The dimensions of the through-holes were measured using a laser confocal microscope (ICP-AES, PerkinElmer, Plasma 400), as shown in Fig. 1(a). The seed layer for the glass through-holes was achieved using a magnetron sputtering process by the Advanced Semiconductor Packaging Research and Development Center Co., Ltd., and the microstructure of the seed layer was observed using a scanning electron microscope (SEM, Gemini SEM 300) equipped with an energy-dispersive spectrometer (EDS), as shown in Fig. 1(b). The through-holes were then metallized using a direct current (DC) electroplating process with a plating solution developed by the research group, at a current density of 0.1 ASD. Following metallization, the glass substrates with through-holes were heat-treated under an inert gas atmosphere. The heat treatment process was set at temperatures of 100 ℃, 200 ℃, and 300 ℃, with a heating rate of 5 ℃/min, maintained at the set temperature for 30 min, and then cooled to room temperature in the furnace. Finally, chemical mechanical polishing was used to remove the surface copper from the heat-treated and original samples, completing the fabrication of the TGV substrate, as shown in Fig. 1(c).

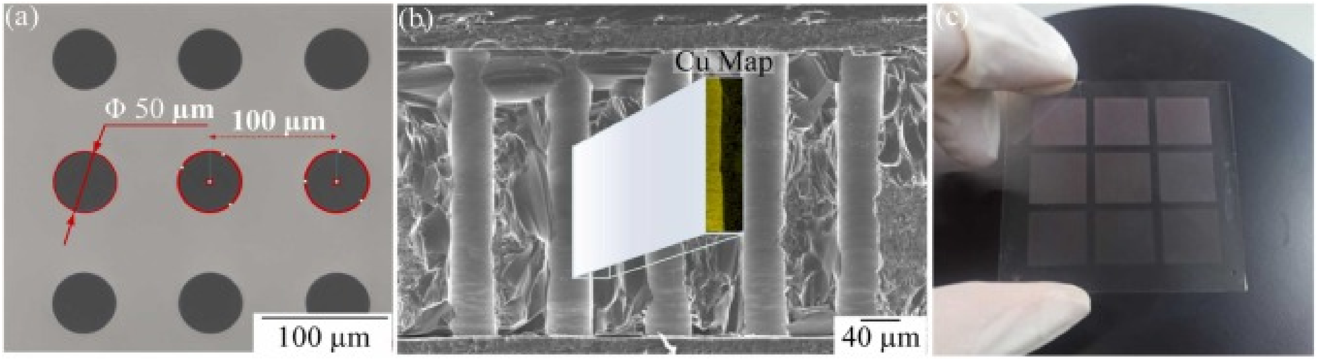

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Fig. 1. (a) The size of original materials, (b) cross-section microstructure and EDS mapping of copper seed layer, (c) the TGV substrate specimen.

In the TGV substrate samples after heat treatment, the microstructure and grain orientation of the copper pillars filling the interconnected through-holes were investigated using EBSD (Gemini 360) technology, with a step size of 0.5 μm and an operating voltage of 20 kV. A high-resolution X-ray 3D imager was used to observe the filling condition of the glass through-holes. SEM and EBSD samples were taken from the cross-sections of the copper pillars in both the heat-treated and original samples. Samples for EBSD observation were prepared using standard metallographic procedures and were subjected to electron polishing, as described in other literature [23]. Additionally, this study employed the four-point probe resistance measurement method (KEYSIGHT, B1500A) to randomly select 5 filled copper pillars from the samples before and after heat treatment, measure their resistivity (ρ), and then calculate the average value. Stress meter (FST, 5000) was used to analyze the evolution of the curvature radius of the TGV substrate after heat treatment. The measurement length and measurement step is 40 mm and 2 mm, respectively. X-ray photoelectron spectroscopy (XPS) (ESCALAB Xi+, Thermo Fisher, USA) was employed to characterize the residual electroplating solution additives during the copper electrodeposition process inside the through-holes. A 4 keV Ar ion beam was used for sputter etching to obtain the depth profiles of the elements. The spatial resolution under the collection conditions was approximately 20 micrometers. Spectrum analysis was conducted using Avantage software developed by Thermo Fisher Scientific Ltd.

The hardness and modulus of the samples were evaluated using instrumental nanoindentation techniques. Nanoindentation tests (Bruker, T1980) were conducted on the plane of the copper pillar section of the polished glass substrate samples. For this purpose, a sharp Berkovich indenter with a tip radius of 100 nanometers was used. The samples were initially ground with SiC sandpaper of different grit sizes (ranging from 100 to 2000 P grit sandpaper), followed by polishing with diamond paste and a colloidal suspension. Well-polished samples were then cleaned with ultrasonic waves to remove any polishing debris from the sample surface. A load control mode with a maximum load (P_{max}) of 40 mN was used in each test, and a dwell time of 10 s was maintained between the loading and unloading processes. To reduce the randomness of the test data, at least 5 indentation tests were performed on each sample. The hardness and modulus of the samples were assessed from the indentation tests.

To discuss the effect of titanium on the Cu-Ti-SiO_2 interface within the through-hole before heat treatment, corresponding density functional theory (DFT) calculations were carried out using the Vienna Ab initio Simulation Package (VASP) code. This work employed the generalized gradient approximation (GGA) through the Perdew-Burke-Eruzerh (PBE) functional. For the interface structure, the cutoff energy was set to 520 eV, with a k-point grid of 9 × 9 × 1, and 12 × 12 × 1 for calculating electronic properties. To eliminate interactions between adjacent layers, a vacuum space of 20 Å was used. The tolerance values after relaxation were set to: total energy convergence below 5×10^{-5} eV/atom. The threshold for ionic forces was set to below 0.01 eV/Å. Van der Waals corrections were applied using the DFT-D3 method with the Becke-Johnson damping function.

3. Results and discussion

3.1. Characterization of manufacturing

Fig. 2 shows scanning electron microscopy images and 3D X-ray renderings of the copper pillars filled in the TGV substrate after electroplating. The parameters of the TGV are listed in Table 1, with a diameter of 50 μm, a height of 300 μm, and a depth-to-diameter ratio of 6:1. The density of the TGV reaches 2×10^4 TGV/mm². As shown in Fig. 2(a), the TGV sidewalls fabricated using laser-induced and wet etching processes are crack-free, with metal uniformly adhered to the sidewalls. The two ends of the TGV are completely covered with metal, without any voids, and there are no defects after thinning, which proves the good reliability of the through-hole metallization process. Since the through-holes are completely filled with metal during the electroplating process, the traditional spin-coating process can still be used for lithography after subsequent chemical mechanical polishing, without the need to add complex manufacturing processes. Fig. 2(b) shows a partial production-scale TGV array, where the filled metal is orderly arranged, and the substrate is free of cracks. Microstructural analysis indicates that the manufactured electroplated metallized TGV substrate has a very high through-hole density and production stability. Therefore, the through-hole electroplating filling process can be used to fabricate metallized TGV substrate.

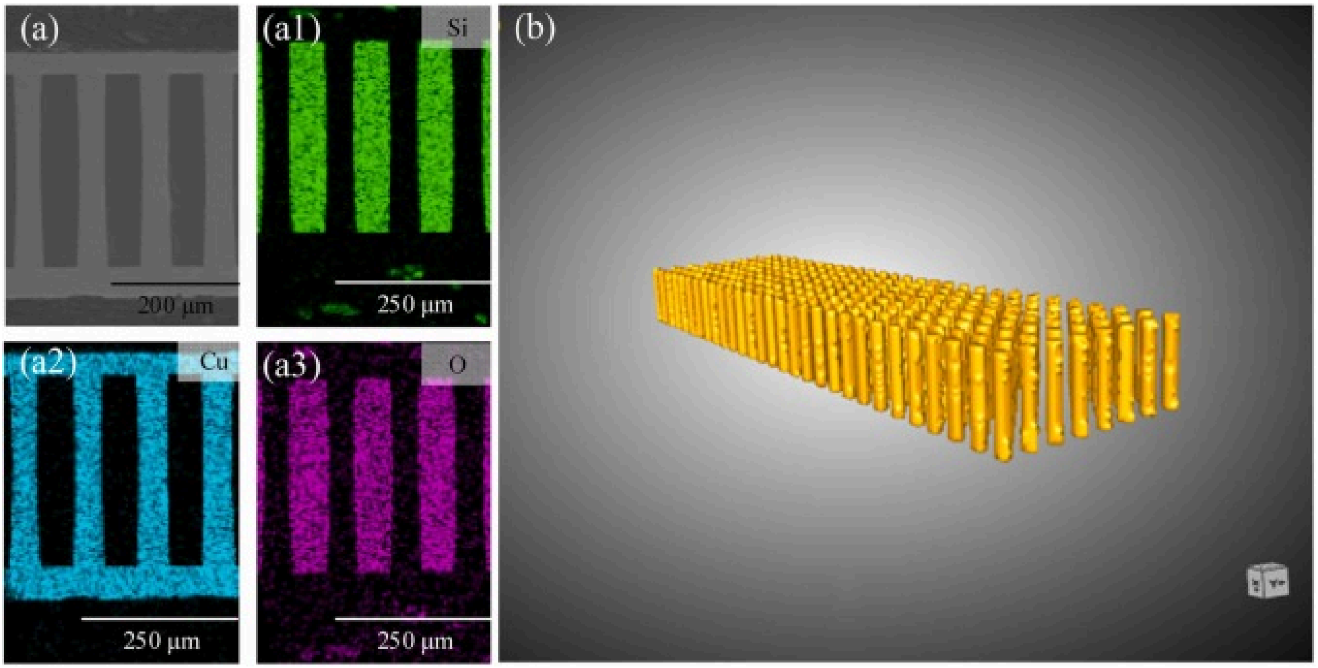

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Fig. 2. (a) SEM image and EDS mappings of cross-section glass substrate and (b) 3D X-ray rendering of the through-hole electroplated copper pillar.

Table 1. TGV array parameters.

Dimension	Numerical value (μm)
Height	300
Diameter	50
The center-to-center distance between two adjacent holes	150

3.2. First-principles calculation

During the heat treatment process, the interfacial bonding strength of the glass (SiO_2)-titanium-copper structure is crucial for the overall performance of the material. Therefore, before implementing heat treatment, we must delve deeply into the influence of the addition of titanium on the interfacial bonding strength of the Cu-Ti-SiO_2 interface. We employ molecular dynamics simulation techniques to conduct an in-depth study of this structure. Through this advanced simulation method, we aim to uncover how titanium atoms act at the interface between quartz and copper, and to explore the microscopic mechanism of interfacial bonding strength evolution. This research not only helps us understand the role of titanium at the quartz/copper interface but also provides a theoretical basis for optimizing heat treatment processes and enhancing the interfacial bonding strength of the glass Cu-Ti-SiO_2 structure.

Fig. 3(a) and (b) show the changes in the distances between atoms before and after relaxation, where **I** and **II** represent the SiO_2-Ti-Cu structures before and after relaxation, respectively. Before and after relaxation, although the distance between silicon and titanium atoms increased from 4.3 Å to 4.6 Å, the distances between silicon and copper atoms, as well as between titanium and copper atoms, decreased from 9.01 Å and 3.1 Å to 7.92 Å and 2.4 Å, respectively. The reason for this result may be that the size of the titanium atom is larger than that of the silicon atom, and during the relaxation process, the coordination number around the titanium atom increases, leading to an increase in the distance between silicon and titanium atoms to reduce steric hindrance, with the titanium atom moving to a more stable but slightly farther position. However, due to the smaller atomic radius of copper atoms compared to that of titanium atoms, copper has a tendency to segregate, and after relaxation, the reduction in constraints on copper atoms allows their distances from silicon and titanium atoms to decrease. To analyze the effect of the changes in atomic distances after relaxation on the bonding between atoms, as shown in Fig. 3(c), the electron localization function of the SiO_2-Ti-Cu structure is presented. This result clearly shows the characteristics of the electron cloud distribution after copper segregation. Notably, in the charge density distribution, the interaction regions between SiO_2 and the titanium element, as well as between the titanium element and the segregated copper element, can be clearly observed. This feature reveals the significant impact of copper segregation on the interface electronic structure and bonding behavior. Moreover, the partial density of states (PDOS) shown in Fig. 3(d) further confirms our findings. In the PDOS of Cu-4s, Ti-4s, Si-3p, and O-2p at the SiO_2-Ti-Cu interface, hybridization peaks as shown in the inset can be clearly observed. These results indicate that the interaction and bonding between SiO_2 and the nearest Ti and Cu atoms are facilitated by the hybridization of Cu-4s, Ti-4s, Si-3p, and O-2p orbitals.

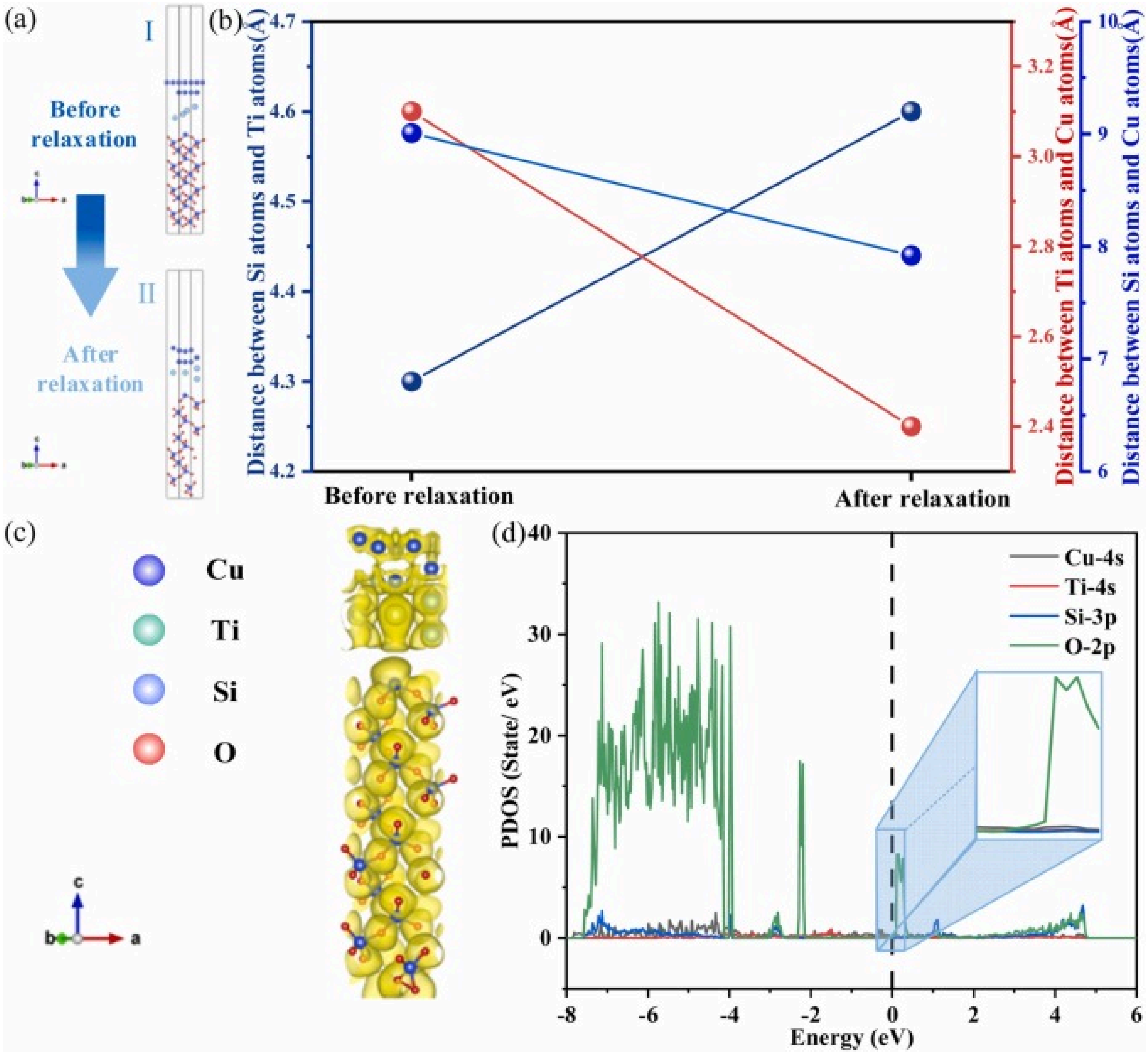

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Fig. 3. SiO_2-Ti-Cu interface structure model. (a) Interface atomic structure of the SiO_2-Ti-Cu system before and after relaxation; (b) atomic distances in the SiO_2-Ti-Cu structural system before and after relaxation; (c) electron localization function of the SiO_2-Ti-Cu structural system; (d) PDOS of the SiO_2-Ti-Cu structural system.

It is noteworthy that theoretical calculations indicate that Cu segregation on the SiO_2-Ti-Cu structure is preferred. This is not surprising, as in the heterogeneous structure of SiO_2-Ti-Cu, the interfacial energy between different elements within the multicomponent system affects their distribution. The interfacial energy between copper and titanium is low [24], [25], leading to a tendency for copper atoms to segregate at these interfaces. Since titanium atoms are deposited on the SiO_2 surface via magnetron sputtering, lattice and chemical potential mismatches result in the SiO_2-Ti interface acting as a source and sink for defect nucleation or annihilation, which is likely the preferred location for Cu segregation. Additionally, due to the large difference in thermal expansion coefficients between the two, high-density defects may be generated in the titanium deposition layer, which provide rapid diffusion channels, such as dislocation glide or vacancy diffusion streams. These defects, when thermally excited during the heat treatment stage, will promote the drag of Cu atoms towards the SiO_2-Ti interface [26], [27]. This Cu segregation phenomenon not only reduces the interfacial energy but also increases interfacial stability and enhances interfacial adhesion, which will have a significant impact on the local dislocation movement at the interface. Furthermore, the enhanced interfacial adhesion achieved through Cu segregation hinders local interfacial sliding and improves the load transfer efficiency of the interface. This significantly contributes to the improvement of the mechanical properties of the SiO_2-Ti-Cu interface.

3.3. EBSD analysis

3.3.1. Grain boundary

Fig. 4 presents comprehensive EBSD results, including IQ (Image Quality) and GOS maps. The results indicate that the copper grains inside the vias are roughly equiaxed. Fig. 4(e) shows the grain size distribution. It can be observed that after heat treatment, the average grain size of the electroplated copper pillars inside the glass vias gradually decreases. The average grain sizes of the as-electroplated, 100 ℃, 200 ℃, and 300 ℃ samples are 3.31 μm, 3.29 μm, 3.16 μm, and 2.86 μm, respectively. Fig. 4(f) shows the distribution of misorientation angles in the cross-sections of the copper pillars inside the vias before and after heat treatment. The calculated average misorientation angles before and after heat treatment are 45.54°, 44.53°, 43.66° and 45.03°, respectively. This indicates that with the increase of heat treatment temperature, the copper pillars inside the vias show an overall trend of first decreasing and then increasing in the distribution of misorientation angles.

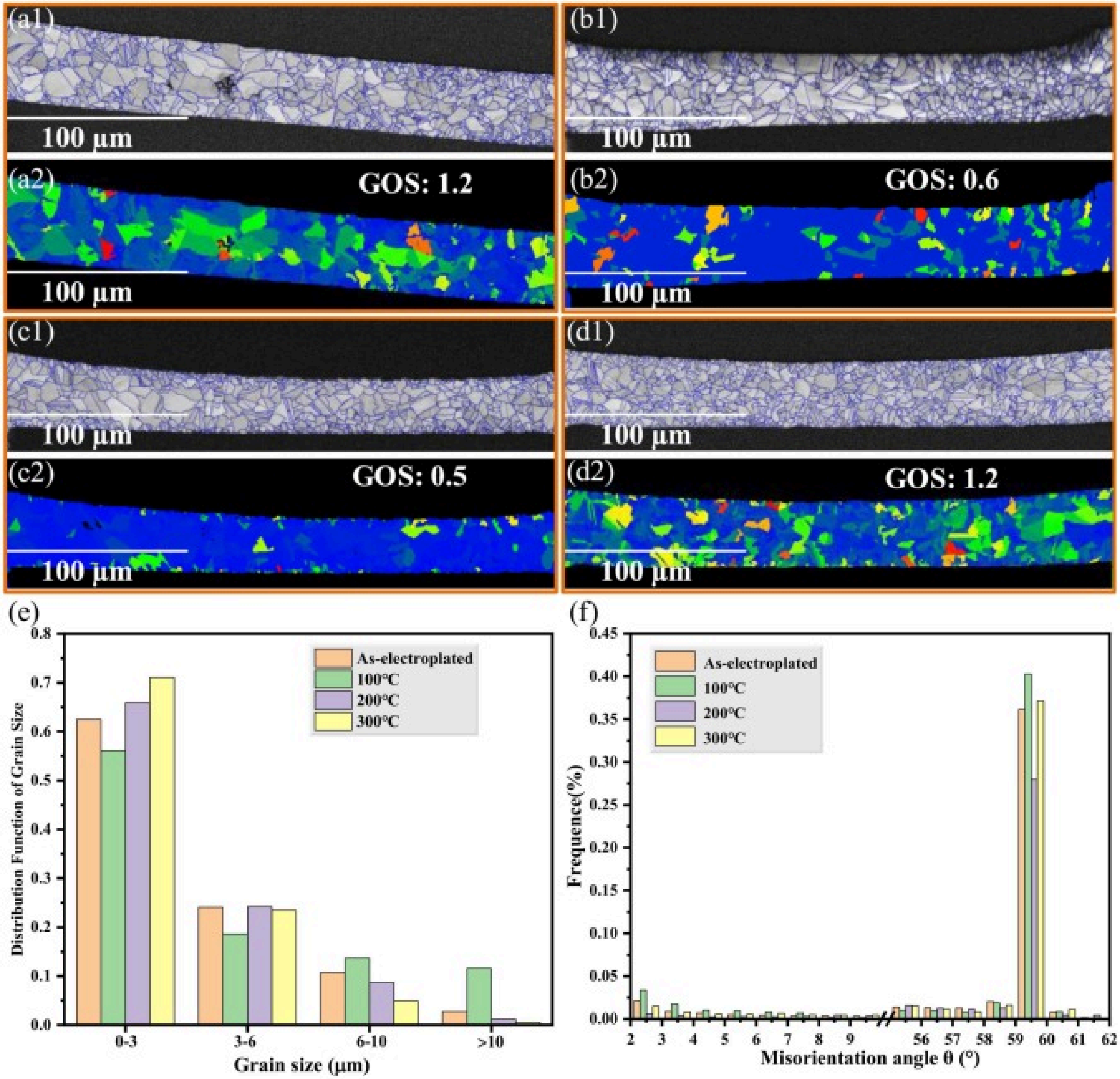

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Fig. 4. Image quality (IQ) maps (a1-d1) and grain orientation spread (GOS) maps (a2-d2); (e) grain size distribution for the samples before and after heat treatment; (f) misorientation angle distribution for the samples before and after heat treatment.

According to the study by Kim et al. [28], when a crack encounters a grain boundary during fracture, it induces a more significant angular distortion. The larger the grain boundary misorientation angle, the higher the distortion and energy consumption associated with crack propagation. Therefore, microstructures with a high density of large-angle grain boundaries (angles greater than 50°) often exhibit more complex and tortuous crack paths during fracture, effectively inhibiting crack propagation [29], [30], [31], [32]. Based on the statistical analysis of the data in Fig. 4(f), the proportions of grain boundary misorientation angles greater than 50° before and after heat treatment are 50.91 %, 51.48 %, 51.63 %, and 43.43 %, respectively. This suggests that at a heat treatment temperature of 200 ℃, the electroplated copper pillars inside the glass vias are more effective in inhibiting crack propagation. The reason for this result may be that, in the initial stage of heat treatment, the dislocation density in the material is high, and the grain size is relatively large. As the temperature increases, the material begins to recrystallize, and the formation of new grains is often accompanied by a larger orientation difference, which may lead to a higher proportion of grain boundaries with misorientation angles greater than 50°. As the temperature continues to rise, the recrystallization process further progresses, and the grains begin to refine. At this stage, due to the migration of grain boundaries and the re-nucleation of grains, the formation of misorientation angles less than 50° increases, resulting in a decrease in the proportion of grain boundaries with misorientation angles greater than 50°.

The heat treatment process affects the ability to inhibit crack propagation on one hand, and on the other hand, due to the significant difference in coefficient of thermal expansion between the copper pillars inside the vias and the glass, the residual stresses and plastic strains generated by the heat treatment process have a non-negligible impact on the reliability of electrical interconnects. GOS analysis is adept at assessing local strain gradients, and the residual stresses and degree of plastic deformation within individual grains of the via-deposited copper pillars before and after heat treatment can be quantified by analyzing GOS [33]. The GOS test calculates the deviation between the average orientation of each grain pixel and the average orientation of its grain [34]. Therefore, a higher GOS value indicates greater distortion within grains, which can easily form crack sources. Grains that are close to the recrystallized state, however, exhibit lower GOS values.

Fig. 4**(a2)** to **(d2)** show the average GOS values of the via-deposited copper pillars before and after heat treatment. The average GOS value of the sample before heat treatment was 1.2. When the heat treatment temperature reached 100 ℃, the average GOS value of the sample decreased to 0.6. As the heat treatment temperature was further increased to 200 ℃, the average GOS value of the sample decreased by 120 % compared to before heat treatment, reaching 0.5. However, when the heat treatment temperature was finally raised to 300 ℃, the average GOS value of the sample increased to 1.2. According to the analysis of the microstructural evolution of the copper pillars inside the vias, the reason for this result may be due to the fact that at the initial stage of heat treatment, the increase in temperature caused dislocations to begin to move, and the interactions between dislocations led to the rearrangement and annihilation of dislocations, thereby reducing the dislocation density of the material. Meanwhile, the movement of dislocations promoted the recrystallization process, and the formation of new grains was often accompanied by a lower dislocation density, as the dislocation density inside recrystallized grains is lower [35]. Due to the formation of recrystallized grains, the grain orientations tended to be uniform, leading to a decrease in GOS values. However, as the heat treatment temperature further increased to 300 ℃, the thermal mismatch between copper and glass led to an increase in internal strain within the grains, and the dislocation density excited was much greater than the dislocation density consumed by rearrangement, thus causing a change in the dispersion of grain orientations and ultimately leading to an increase in GOS values. This will exacerbate stress concentration within the deposited copper pillars inside the vias and induce crack propagation.

In summary, as the heat treatment temperature increases, the average grain size of the deposited copper pillars inside the vias gradually decreases. The increased number of grain boundaries and the structural disorder inhibit dislocation slip, resulting in grain boundary strengthening. However, the degree of lattice distortion within the copper pillars shows a trend of first decreasing and then increasing. After heat treatment at 200 ℃, the lattice distortion within the copper pillars is minimized, which helps to reduce local stress concentration and maximizes its resistance to crack propagation.

3.3.2. Coincidence site lattice analysis

When discussing the characteristics of grain boundaries, the coincidence site lattice (CSL) is also a key element that deserves attention, as it significantly impacts the mechanical properties of polycrystalline materials. At certain specific misorientation angles, certain lattice points within a grain will completely align with the lattice points of the adjacent grain. Thus, the CSL is composed of the sites (atoms) where the lattices of two neighboring grains coincide [36]. In the field of research on polycrystalline materials, grain boundaries can prevent the movement of dislocations, but the capacity of grain boundaries to accommodate dislocations is limited, leading to an increase in material strength while ductility decreases. Although the slip process in polycrystals is significantly affected by general grain boundaries and special grain boundaries (such as twin boundaries (TB)), TB typically exhibits higher thermal stability and mechanical stability compared to traditional high-angle grain boundaries (GB). Twin boundaries are a special type of CSL grain boundary with a symmetry of mirror lattice at the boundary, meaning that the atoms on one side of the boundary are exactly at the mirror position of the atoms on the other side [37]. In the academic literature, twin boundaries are often specifically mentioned [38], [39], [40], [41] because their high density can confer greater strength to alloys. Researchers such as Zhu et al. [42] and Meng et al. [43] have pointed out that CSL boundaries can effectively block the passage of dislocations, thereby influencing plastic deformation. Studies have shown that the Σ-3 twin boundary [44] provides a critical energy barrier for Cu with low stacking fault energy, preventing dislocation slip between twins, thereby giving the material higher yield stress[45], [46].

The classification of CSL is done using the sigma (Σ) notation, which indicates the density of coincident lattice points in the reciprocal space [47]. The Σn type defines a CSL where one atom out of every ‘n’ atoms is shared by two adjacent lattices. For FCC polycrystalline copper with an m3m symmetric space group, the primary types of CSL boundaries are Σ3 (<111 >60°, TB), Σ5 (<110 >36.9°), Σ7 (<111 > 38.2°), and Σ9 (<110 >38.9°, TB). Studies have found that the evolution and distribution of CSL grain boundaries are influenced by processing conditions. Therefore, under different heat treatment processes, the types and percentages of CSL boundaries in electroplated copper pillars within vias should be significantly different. Fig. 5(a) shows the percentage of Σ3, Σ5, Σ7, and Σ9 in the samples before and after heat treatment, as determined by EBSD analysis. For each sample, the proportion of Σ3 boundaries is the highest. They occur most frequently because their energy is lower than that of other boundaries. In this experiment, Σ9 boundaries are the second most common type of boundary.

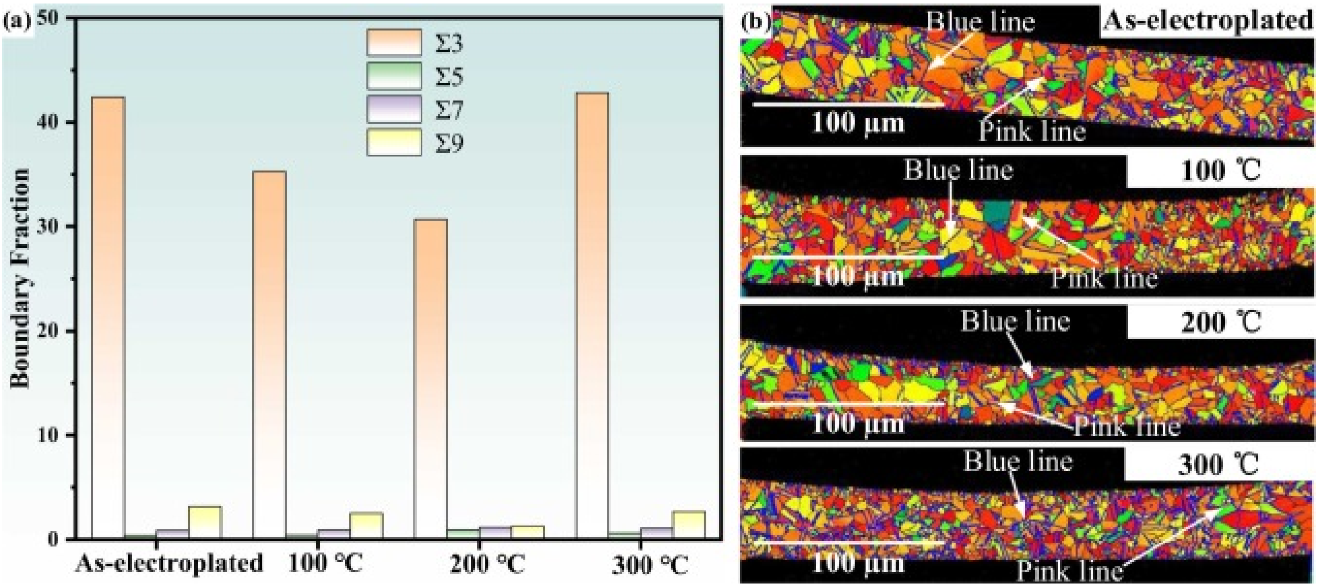

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Fig. 5. (a) Fraction of CSL boundaries of through-hole electroplated copper pillar before and after heat treatment; (b) schmid factors and Σ3-TB (blue lines) and Σ9-TB (pink lines) images of through-hole electroplated copper pillar before and after heat treatment.

For the sum of the proportions of Σ3 and Σ9 twin boundaries, when the heat treatment temperature is increased from room temperature to 100 °C, their proportion decreases from 45.56 % to 37.78 %, and when the heat treatment temperature is further increased to 200 °C, their proportion further drops to 31.97 %. For polycrystals with a high content of Σ3 boundaries, under the action of thermal activation, Σ3 boundaries tend to form Σ3-Σ3-Σ9 triple junctions [48], reducing the grain boundary energy through orientation changes [49]. Therefore, in this study, it is reasonable that the proportions of Σ3 and Σ9 boundaries show the same trend with the increase of heat treatment temperature. However, surprisingly, when the temperature is increased to 300 °C, the proportion of both does not decrease but instead rises to 45.46 %.

Fig. 5(b) shows the Schmid factor and Σ3 twin boundary (blue line), Σ9 twin boundary (pink line) images of electrodeposited via copper pillars under different heat treatment temperatures, where the Schmid factor map is superimposed with Σ3 and Σ9 type

TBs for visualization. It is well known that under suitable conditions, mechanical tensile stress can induce twin formation in polycrystalline copper. Combined with the GOS analysis in Section 3.3.1, after the heat treatment temperature is increased from 200 °C to 300 °C, due to the large difference in thermal expansion coefficients between glass and copper, the stress within the constrained electrodeposited copper pillars shifts from a relaxed state to an activated state, resulting in an increase in the proportion of Σ3 and Σ9 twin boundaries. To test whether the combined thermal stress force in the circumferential direction can act as mechanical tensile stress, we employed the Schmid's law. Based on the {111}< 11−2 > twin system of face-centered cubic (fcc) metals, the schmid factor for copper grains under the action of the combined constrained thermal stress force can be derived: according to statistical results, Σ3 and Σ9 type twin boundaries primarily appear in areas with higher Schmid factors [50]. Since the critical resolved shear stress (CRSS) is constant, the higher the Schmid factor, the lower the external applied stress required to initiate twin deformation. Based on the Schmid factor analysis, a brief conclusion can be drawn: under the action of combined constrained thermal stress, grains within the copper pillars that require lower necessary applied stresses are more likely to initiate twin deformation, thereby inducing the formation of twin boundaries.

However, the direction of the combined force is not so straightforward and has not yet been clearly defined. Fig. 6(a) to (d) show the inverse pole figure (IPF) of the copper pillar within the via before and after heat treatment. The X and Z axes represent the radial inverse pole figure, while the Y axis represents the axial inverse pole figure. Although the copper pillar experiences complex stresses in microregions, its radial inverse pole figure consistently shows a (101) preferred orientation, which suggests that the stress in the circumferential direction is in a state of equilibrium. However, along the axial direction, the preferred orientation of the copper pillar is changing. Before heat treatment, both the (101) and (111) directions show clear preferred orientations, but after the temperature is increased to 100 °C, the preferred orientation of the copper pillar becomes (111). As the temperature increases further, at 200 °C, the preferred orientation is (001). When the heat treatment temperature finally reaches 300 °C, the preferred orientation of the copper pillar is (101). This indicates that during the heating process, the direction of the constrained thermal stress is along the axis of the via, causing the grains of the copper pillar to rotate, thus exhibiting different preferred orientations, as shown in the schematic diagram in Fig. 6(e). The twin deformation induced by the combined constrained thermal stress suggests that under the action of thermal stress, the strain behavior of copper in a cylindrical confined space can be analogous to the external tensile strain in traditional solid mechanics.

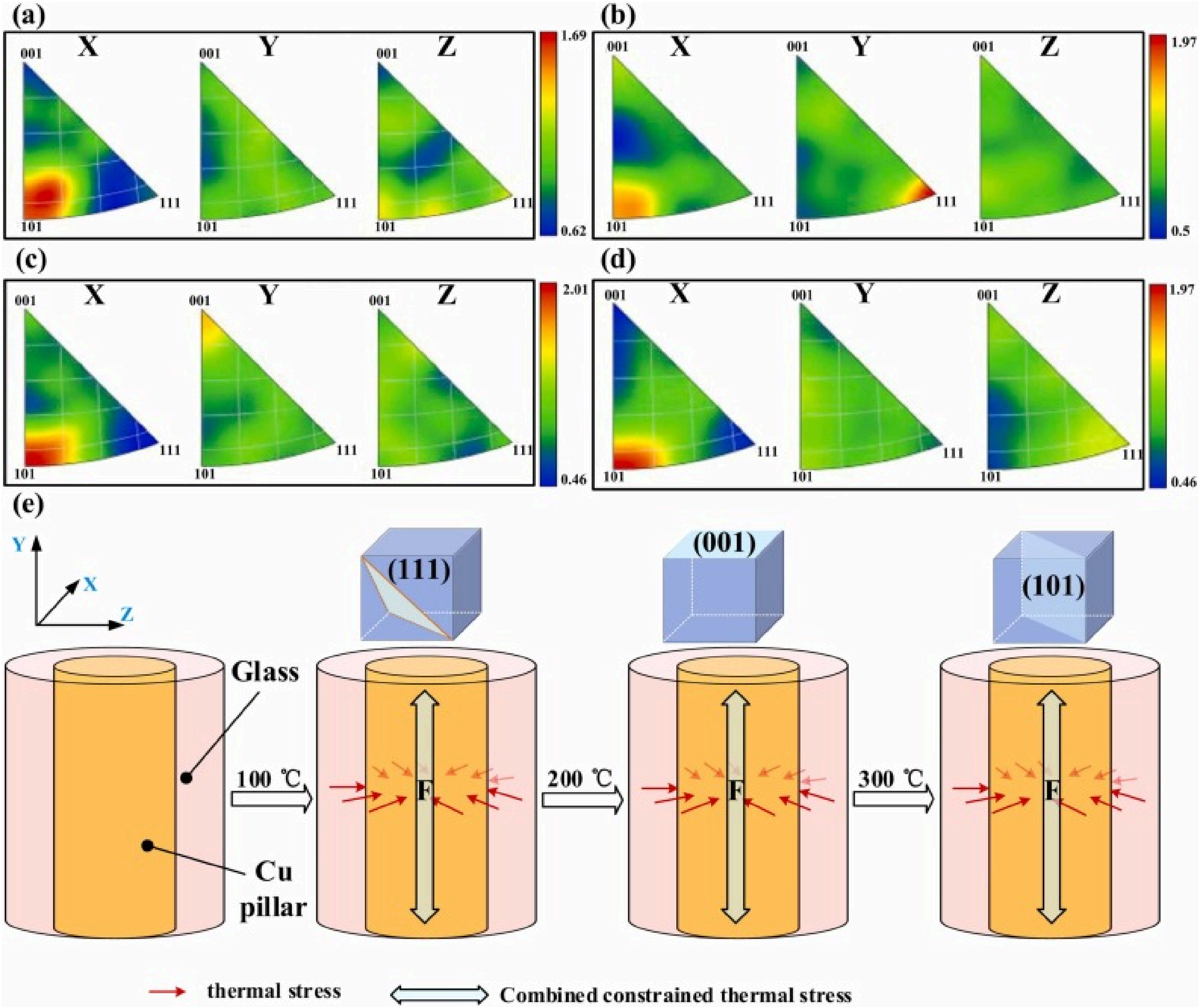

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Fig. 6. IPF maps of the specimens (a) as-electroplated; (b) 100 ℃; (c) 200 ℃; (d) 300 ℃; (e) constrained thermal stress direction diagram.

3.4. Nano-indentation behavior

In the microelectronics manufacturing industry, the TGV substrate in chip packaging serves as a key component that facilitates vertical connections, and its processing quality directly affects the performance and reliability of the entire microelectronics system. Generally speaking, the behavior and tribological properties of copper pillars during the grinding and polishing process are often influenced by the inherent characteristics of the material itself. Against this backdrop, the impact of heat treatment on the grinding and polishing process of copper pillars cannot be ignored. Therefore, this chapter aims to explore the micro-nano mechanical behavior and changes in tribological properties of copper pillars within vias before and after heat treatment, with the intention of providing theoretical basis and empirical support for optimizing the grinding and polishing process and improving the quality of TGV substrates during grinding and polishing.

Fig. 7(a) shows the corresponding load-displacement curves obtained from nanoindentation experiments conducted on the cross-section of the copper pillars before and after heat treatment, with the corresponding hardness (H), elastic modulus (E), and maximum indentation depth (d_{max}) illustrated in Fig. 7(b). It can be noted that as the heat treatment temperature increases, the maximum indentation depths of the copper pillars within the vias at room temperature, 100 ℃, 200 ℃, and 300 ℃ are 1130 nm, 1255 nm, 1108 nm, and 1060 nm, respectively, showing a trend of first increasing and then decreasing. The decrease in d_{max} indicates that the hardness of the copper pillars has been enhanced.

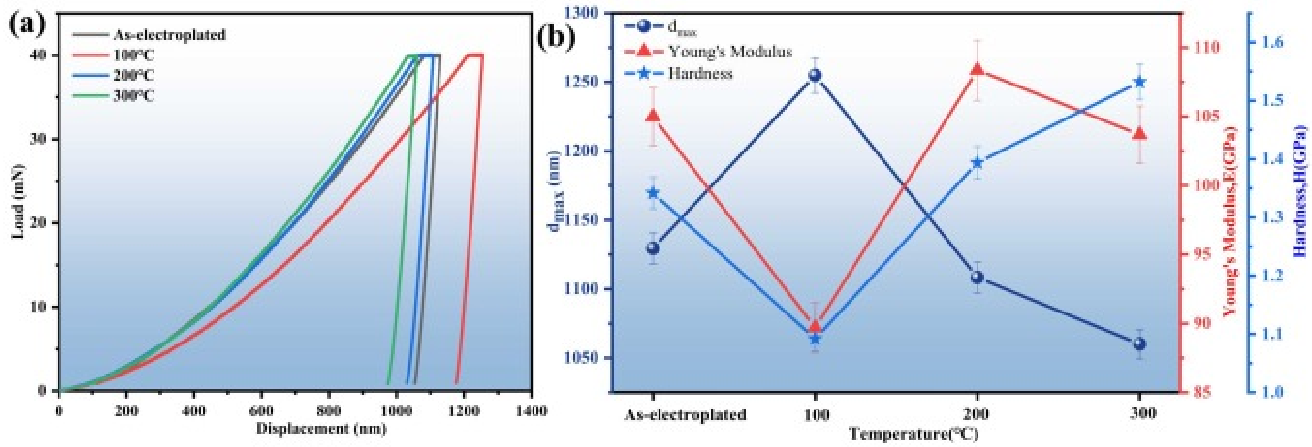

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Fig. 7. (a) Nanoindentation load-displacement curves of through-hole electroplated copper pillars before and after heat treatment; (b) d_{max}, Young's modulus (E) and hardness (H) of through-hole electroplated copper pillars before and after heat treatment.

Fig. 7(b) clearly verifies the trend in the nano-hardness of the copper pillars. The hardness of the copper pillars at room temperature is 1.34 GPa. When the heat treatment temperature reaches 100 ℃, the hardness of the copper pillars is 1.09 GPa, which is a 19 % decrease from the room temperature state. As the temperature further increases to 200 ℃, the hardness of the copper pillars reaches 1.39 GPa, slightly exceeding the hardness at room temperature. Finally, when heat treated at 300 ℃, the hardness of the copper pillars is 1.53 GPa, an increase of 14 % compared to the room temperature state. According to the microstructural analysis in Section 3.3.1, the nano-hardness of the electro-deposited copper pillars within the glass vias before and after heat treatment shows the aforementioned trend due to the combined effect of grain refinement and lattice distortion. The hardness of copper pillars is primarily influenced by the combined effect of their internal average grain size and the degree of lattice distortion. Specifically, the smaller the average grain size, the greater the number of grain boundaries, which in turn strengthens the hindrance to dislocation slip, resulting in a higher hardness value of the copper pillars in nanoindentation tests. The degree of lattice distortion also has a positive correlation with hardness. When the temperature rises from room temperature to 100 ℃, although the average grain size of the copper pillars slightly decreases, the internal GOS value decreases by 50 %. The effect of reduced lattice distortion promoting a decrease in hardness far outweighs the effect of grain refinement promoting an increase in hardness. However, as the heat treatment temperature increases to 200 ℃, although both the average grain size and GOS value of the copper pillars continue to decrease, the effect of reduced lattice distortion promoting a decrease in hardness is weaker than the effect of grain refinement promoting an increase in hardness. When the heat treatment temperature increases to 300 ℃, the decrease in the average grain size of the copper pillars and the increase in the GOS value together contribute to the improvement of the nano-hardness of the copper pillars.

Additionally, the nanoindentation mechanical behavior of the copper pillars can provide valuable insights into their wear resistance. The elastic modulus plays a crucial role in determining the wear behavior of materials. The wear resistance can be assessed by evaluating the ratio of nano-hardness to elastic modulus. The parameter H/E represents the maximum load-bearing capacity of a material while maintaining elastic behavior during friction. A higher H/E value indicates better wear resistance. Similarly, the parameter H^3/E^2 is also important in assessing the resistance of material to plastic deformation [51], [52]. In this regard, the universal hardness (HU) is a key factor affecting the material's anti-wear capability and can be determined by considering both elastic and plastic components [53], [54]:

$$HU = \frac{F_{\max}}{(26.43 * d_{\max})^2}$$

Among them, F_{max} is the maximum load stress during nanoindentation (40 mN). The universal hardness (HU) of the electroplated copper pillars before and after heat treatment, as well as the ratio of H/E, are listed in Table 2. Notably, the samples heat-treated at 300 ℃ exhibit higher HU and H/E, indicating better wear resistance compared to the other samples. It is important to emphasize that during the thinning stage of the TGV substrate, the tribological properties required for the removal of the copper film on the glass surface are different. Due to the significant difference in material hardness between copper and glass, a higher HU value results in a smaller depth of the concave copper pillars after thinning, which helps to improve the yield of mass production. However, the effect of heat treatment at 300 ℃ on the warpage curvature radius of the electroplated metallized TGV substrate is not yet clear, so warpage analysis will be conducted next.

Table 2. Important parameter for measuring the wear resistance of metal materials.

Sample	H/E($\times10^{-2}$)	H^3/E^2($\times10^{-4}$ GPa)	HU($\times10^{-2}$ GPa)
As-electroplated	1.28	2.19	4.49

Sample	H/E($\times10^{-2}$)	H^3/E^2($\times10^{-4}$ GPa)	HU($\times10^{-2}$ GPa)
100 ℃	1.22	1.62	3.64
200 ℃	1.29	2.31	4.66
300 ℃	1.48	3.34	5.10

3.5. Warping analysis

Histogram of the warpage curvature radius of the copper overburden TGV substrate before and after heat treatment is shown in Fig. 8. With the help of the fitting program provided by the testing equipment, the fitted warpage radius of the copper electroplated TGV substrate decreased first and then increased as the heat treatment temperature was raised from room temperature to 300 ℃. The fitted warpage radius at room temperature was 766 m, and when the temperature was raised to 100 ℃, the fitted warpage radius of the copper electroplated TGV substrate was 979 m. As the heat treatment temperature continued to rise, after heat treatment at 200 ℃, the fitted warpage radius decreased to 687 m. Finally, after heat treatment at 300 ℃, the fitted warpage radius of the copper electroplated TGV substrate decreased to 498 m. This means that the fitted warpage radius was the largest at a heat treatment temperature of 100 ℃, indicating the smallest warpage height. The warpage radius was the smallest at a heat treatment temperature of 300 ℃, indicating the largest warpage height.

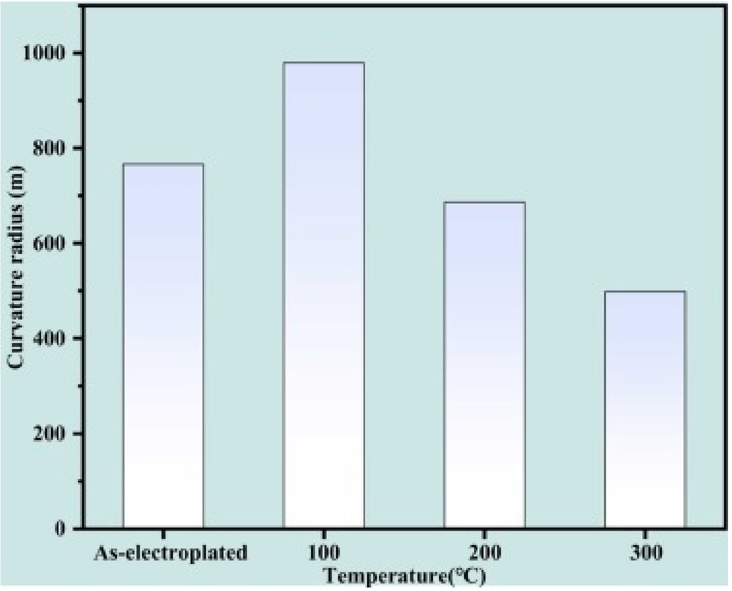

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Fig. 8. Histogram of warpage curvature radius of TGV substrate before and after heat treatment.

Under normal circumstances, an increase or decrease in temperature leads to a nonlinear change in wafer curvature, primarily due to the transformation of grain structure or plastic deformation during the heat treatment process, which causes inelastic deformation. The inelastic deformation during heat treatment results in the generation of residual stresses (σ_r) after cooling to room temperature (RT), subsequently causing wafer warpage. This phenomenon can be elucidated by the provided Stoney formula [55].

$$\sigma_r = E_s h_s^2 (K_f - K_s)/6h_f(1 - \upsilon_s)$$

In this formula, E_s and h_s represent the elastic modulus and thickness of the substrate, υ_s is the Poisson's ratio of the substrate, h_f is the thickness of the film, K_s is the curvature of the substrate, and K_f is the curvature of the film. According to the above formula and our former research [18], when the warpage fitting radius is at its minimum at 100 °C, the absolute value of the residual stress at this temperature should also be the smallest. However, interestingly, based on the previous calculation of the internal residual stress (σ_{XRD}) in the copper layer on the glass surface after heat treatment using the $\sin^2\Psi$ method, the absolute value of the residual stress shows a trend of first decreasing and then increasing with the increase of heat treatment temperature. The absolute value of the residual stress is the smallest at a heat treatment temperature of 200 ℃. The reasons for the differences in the above results are as follows:

Firstly, σ_{XRD} represents the static equilibrium deformation in the crystalline portion characterized by the $\sin^2\Psi$ method, while σ_r calculated by the curvature method is not limited to the macroscopic displacement of crystalline materials. Therefore, in terms of the source of the average microstress on the surface copper layer calculated, it is inherently different from that calculated by the curvature method. For example, when calculating the residual stress caused by the mismatch of the thermal expansion coefficients between the film and the substrate in amorphous or polymer material systems, it can be easily measured by the curvature method, but the $\sin^2\Psi$ method based on X-ray diffraction (XRD) cannot be used for calculation.

Secondly, in terms of the calculation formula for the residual stress of the copper overburden film, the σ_{XRD} calculated by the $\sin^2\Psi$ method incorporates the elastic modulus of the copper-coated film at different temperatures. Since all parameters, except for the Young's modulus of copper, can be considered constant during the heat treatment process, the Young's modulus of copper will vary with temperature and deformation. Therefore, the trend in the absolute value of σ_{XRD} will also be influenced to some extent by the changing trend of the elastic modulus of the copper overburden film. In contrast, the calculation of σ_r using the curvature method primarily judges the trend of residual stress in the single-layer film structure based on changes in the warping fitting radius. For TGV substrates using double-sided simultaneous electroplating, the Stoney formula is not applicable for calculating residual stress.

Therefore, based on the comprehensive comparative analysis above, on one hand, for samples with varying elastic modulus, the residual stress results calculated using the $\sin^2\Psi$ method are more reliable. On the other hand, in this study, the reasons for the changes in the warping curvature radius of the copper overburden glass substrate cannot be explained by the Stoney formula. The warping curvature radius of the copper overburden glass substrate is related not only to the residual stress of the copper layer but also to the elastic modulus of the copper layer.

3.6. Electrical property

This study also focuses on the analysis of the electrical performance of the interconnect copper pillars. Fig. 9 shows the variation curve of the electrical resistivity of the electroplated copper pillars in the through-hole before and after heat treatment. According to the data changes in the figure, it can be seen that the resistivity of the copper pillars increases with the rise in heat treatment temperature. The resistivity value increases from 15.83 × 10^{-6} Ω·m at room temperature to 18.11 × 10^{-6} Ω·m after heat treatment at 300℃. Generally speaking, the microstructure of a material has a significant impact on its resistivity. As indicated in Section 3.3.1, the average grain size of the electroplated copper in the through-hole gradually decreases, which will affect the resistivity in several ways. On one hand, the number of grain boundaries increases. When electrons pass through grain boundaries, scattering occurs, which increases the frequency of electron scattering, thereby reducing the mean free path of electrons and leading to an increase in resistivity [56]. On the other hand, when the grain size decreases, the proportion of vacancy atoms and crystal defects in the material relatively increases, and these point defects will also scatter electrons, reducing electron mobility and resulting in an increase in resistivity [57], [58].

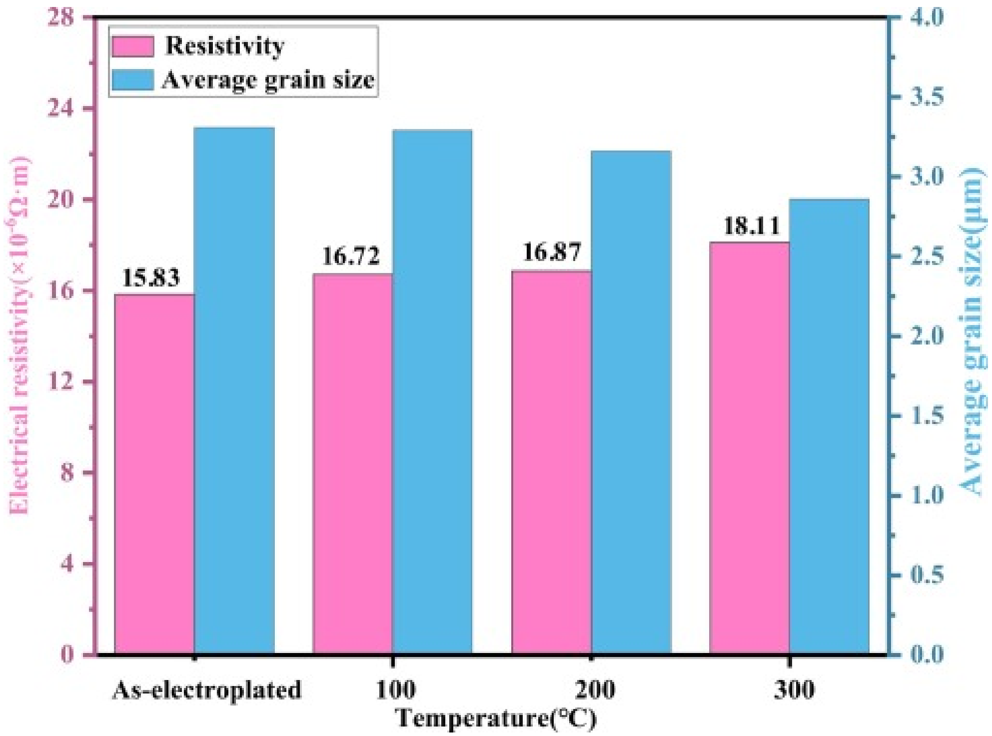

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Fig. 9. Histogram of electrical resistivity and average grain size of through-hole electroplated copper pillars before and after heat treatment.

However, curiously, according to the study by Sun et al. [59], [60], for oxygen-free copper, the resistivity caused by grain boundaries is in the order of 10^{-8}, while the resistivity caused by dislocations is usually in the order of 10^{-10}. The resistivity of the electroplated copper measured in this study is in the order of 10^{-6}. There is a significant difference between the two, indicating that there must be other factors affecting the change in resistivity. Considering that various electroplating additives are used during the generation of electroplated copper, which is vastly different from the production process of oxygen-free copper, the chemical composition of the electroplated copper will be analyzed next.

XPS testing technology was used to characterize the impurity elements in electroplated copper at different heat treatment temperatures. The field of view diameter was 500 micrometers. Prior to testing, Ar ion beam etching was used to remove surface adsorbed C and O. Fig. 10 shows the XPS survey spectra of electroplated copper and calibration copper after Ar ion beam etching at different heat treatment temperatures. The atomic percentage of elements is listed in Table 3. For all electroplated copper samples, the presence of carbon is observed, and the atomic percentage of carbon is significantly higher than that of copper. Based on the process conditions of sample preparation, such a high atomic percentage of carbon primarily originates from various macromolecular organic additives in the electroplating solution. This clearly indicates that during the electroplating process of copper, these additives, due to insufficient time to escape, ultimately remain in the electroplated copper. During the grain refinement process, carbon atoms in the carbon-containing organics may be attracted to the high-energy grain boundaries, leading to segregation at the grain boundaries and the formation of tiny insulating regions. These insulating regions block the flow path of electrons, increase resistance, and cause a decrease in electrical conductivity. Therefore, the residual macromolecular organic additives affect the electrical conductivity of copper, which may be the reason for the higher resistivity of electroplated copper compared to oxygen-free copper.

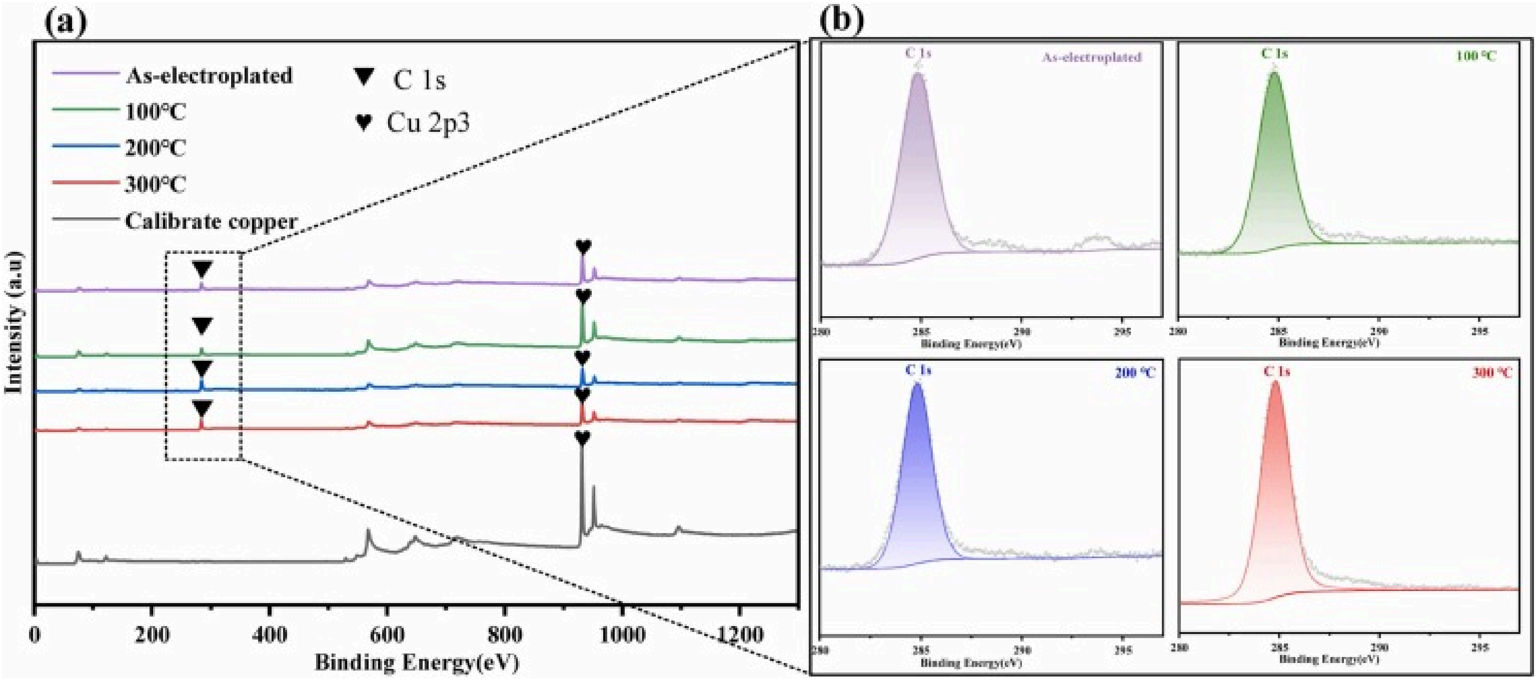

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Fig. 10. (a) Survey XPS spectra of the cu electrodeposited at different temperature; (b) high resolution spectra of C1s for cu electrodeposited at different temperature.

Table 3. The composition of electroplated copper at different heat treatment temperatures.

Element	Binding energy (eV)	Atom percentage (at%)			
		As-electroplated	100℃	200℃	300℃
Cu 2p3	933	19	28	11	14
C 1 s	285	72	66	84	80

Overall, after heat treatment, significant changes occur in the microstructure, mechanical properties, and electrical performance of the electroplated copper pillars within the through holes. After heat treatment at 200 ℃, the lattice distortion is minimized, which is beneficial for inhibiting crack propagation. At 300 ℃ heat treatment, the average grain size is minimized, and due to the constraint thermal stress-induced twin strengthening effect, the wear resistance of the electroplated copper pillars is improved, which helps to reduce the concave depth on the surface of the copper pillars after grinding. However, the minimum warpage height, which is more critical for the TGV substrate, occurs after heat treatment at 100 ℃. Additionally, the lattice distortion of the electroplated copper pillars after 100 ℃ heat treatment is only slightly greater than that after 200 ℃ heat treatment, and the resistivity is lower than that after 200 ℃ heat treatment. Therefore, after considering the warpage height of the TGV substrate, the microstructure, mechanical properties, and electrical performance of the copper pillars, heat treatment at 100 ℃ is more suitable to achieve excellent comprehensive mechanical and electrical performance. Fig. 11 is an image of wafer-level bonding of the electroplated metallized TGV substrate after processing.

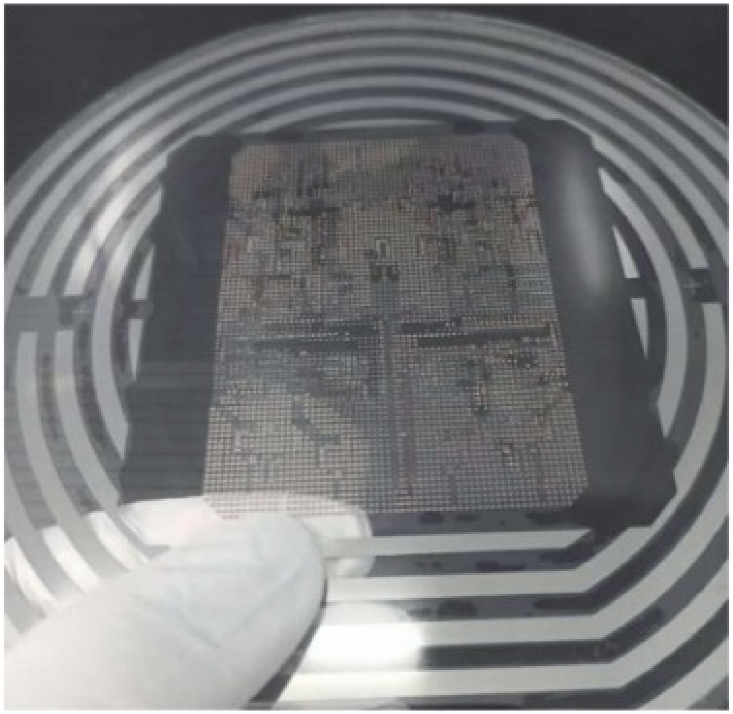

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Fig. 11. Bonded electroplated metallized TGV substrate after processing.

4. Conclusion

In this study, low warpage electroplated metallized TGV substrates were successfully fabricated. We systematically investigated the evolution of the microstructure of vertical interconnect copper pillars by EBSD, as well as the microscopic mechanisms behind the changes in micro-nano mechanical and electrical properties, and deeply analyzed the correlation between the changes in the warpage curvature radius of the TGV substrates and the residual stress of the copper overburden film. The following are some main conclusions:

(1) Titanium atoms have a significant impact on the SiO_2-Ti-Cu structural interface area. The presence of titanium atoms promotes the segregation of Cu at the interface, increasing interface stability and enhancing interfacial adhesion. After heat treatment, due to interatomic diffusion, the bonding strength between copper and SiO_2 at the interface will be further enhanced.

(2) EBSD analysis indicates that during heat treated at 200 ℃, the copper pillars within the glass via have less stress concentration and can more effectively inhibit crack propagation. However, the proportion of Σ3 and Σ9 twin boundaries is the smallest at this temperature. As the temperature rises from 200 ℃ to 300 ℃, a constrained thermal stress-induced twinning phenomenon similar to the mechanical stress-induced twinning occurs, and the proportion of twin boundaries begins to increase.

(3) During the heat treatment process, the warpage radius of the TGV substrates is the smallest after heat treatment at 100 ℃. However, due to the differences in the principles of calculating residual stress by the curvature method and the $\sin^2\Psi$ method, for copper-overburden TGV substrates, the residual stress results calculated by the $\sin^2\Psi$ method are more reliable.

(4) Due to the combined effects of average grain size, lattice distortion, and residual electroplating additives during the heat treatment process, the micro-nano mechanical properties and electrical interconnect performance of the electroplated copper pillars within the via show different trends. Considering the performance requirements of the electroplated metallized TGV substrates for low warpage and the mechanical and electrical properties of the interconnect copper pillars, heat treatment at 100 ℃ is the optimal choice.

CRediT authorship contribution statement

Miao Wang: Investigation, Methodology, Formal analysis, Writing– original draft, Writing– review & editing. **Jihua Zhang:** Resources, Project administration, Funding acquisition. **Libin Gao:** Resources, Project administration. **Hongwei Chen:** Resources, Project administration. **Wenlei Li:** Investigation, Methodology. **Juefeng Fu:** Formal analysis. **Lei Zhao:** Formal analysis. **Yangxu Qiao:** Formal analysis. **Ke Qin:** Formal analysis. **Dongbin Wang:** Resources, Project administration. **Shuang Li:** Resources, Project administration. **Ting Liu:** Resources, Project administration. **Xingzhou Cai:** Formal analysis. **Yong Li:** Formal analysis. **Bin Peng:** Data curation. **Wanli Zhang:** Resources, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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